

Programming Assignment Report

Design and Analysis of Algorithms

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Problem Name: Angry Professor

Problem Statement: A Discrete Mathematics professor has a class of students. Frustrated with their lack of discipline, the professor decides to cancel class if fewer than some number of students are present when class starts. Arrival times go from on time (arrivalTime $\leq$ 0) to arrived late (arrivalTime $>$ 0).

Given the arrival time of each student and a threshold number of attendees, determine if the class is cancelled.

Problem HackerRank Link: <https://www.hackerrank.com/challenges/angry-professor/problem>

Github Submission Link: <https://github.com/Aarushi0403/DAA-LAB-/blob/main/Hackerrank%20problem>

Solution Steps:

- **Start** the program.
- **Input** the number of test cases t.
- **Repeat** the following steps for each test case until t becomes 0:
 1. Read the total number of students n.
 2. Read the threshold value k (minimum students required to start the class).
 3. Create a list a to store the arrival times of students.
 4. Input the arrival time of n students and store them in the list.
 5. Initialize a variable onTimeCount = 0.
 6. **For each** arrival time in the list:
 - If arrivalTime $\leq$ 0, increase onTimeCount by 1.
 7. Check the condition:
 - If onTimeCount $<$ k, print "YES" (class is cancelled).
 - Otherwise, print "NO" (class is not cancelled).

- **End** the loop when all test cases are completed.
- **Stop** the program

Complete JAVA Code:

```
import java.io.*;
import java.util.*;
public class Solution {
    public static String angryProfessor(int k, List<Integer> a) {
        int onTimeCount = 0;
        // Count students who arrived on time (arrival time <= 0)
        for (int arrivalTime : a) {
            if (arrivalTime <= 0) {
                onTimeCount++;
            }
        }
        // If on-time students are less than the threshold k, cancel class (YES)
        if (onTimeCount < k) {
            return "YES";
        } else {
            return "NO";
        }
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int t = scanner.nextInt(); // Number of test cases
        while (t-- > 0) {
            int n = scanner.nextInt(); // Total students
            int k = scanner.nextInt(); // Threshold
            List<Integer> a = new ArrayList<>();
            for (int i = 0; i < n; i++) {
```

```

        a.add(scanner.nextInt());
    }
    System.out.println(angryProfessor(k, a));
}
scanner.close();
}
}

```

Observation/Solution:

- The program checks whether a class should be cancelled based on student arrival times.
- A student is considered **on time** if their arrival time is **less than or equal to 0**.
- The program counts the number of students who arrived on time.
- The value **k** represents the **minimum number of students required** for the class to continue.
- If the number of on-time students is **less than k**, the class is **cancelled**.
- If the number of on-time students is **greater than or equal to k**, the class **continues**.
- The program repeats this process for **multiple test cases** and prints "YES" if the class is cancelled, otherwise "NO".